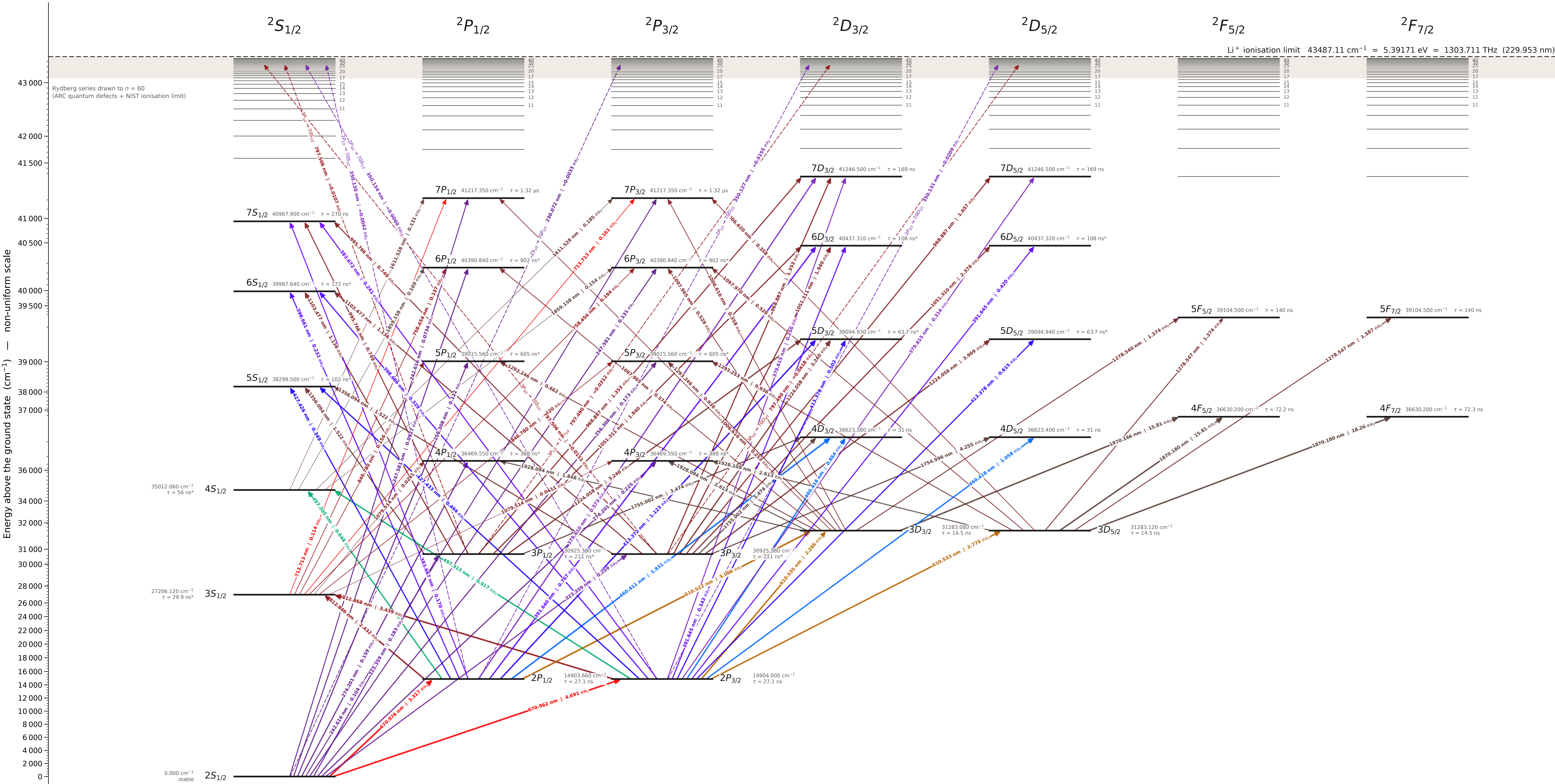
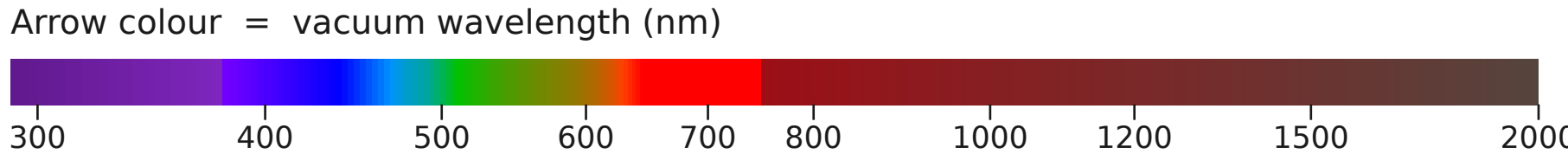


Lithium-7

⁷Li (Li II), nuclear spin *I* = 3/2. Energy levels and transitions below 2 μm: 32 levels, 85 lines.

Arrow colour = vacuum wavelength (nm) | reduced dipole matrix element $\langle J||er||J' \rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked ≈).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$\langle J er J' \rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Typical use
2P _{1/2} – 3D _{3/2}	610.5222	610.3533	491.0427	16379.420	5.066	5.714e+07	10.980	83.3	
2S _{1/2} – 2P _{3/2}	670.9616	670.7764	446.8102	14903.983	4.691	3.690e+07	5.872	100.0	D2 line
2S _{1/2} – 2P _{1/2}	670.9761	670.7909	446.8005	14903.660	3.317	3.690e+07	5.872	100.0	D1 line
2P _{3/2} – 3S _{1/2}	812.8680	812.6446	368.8083	12302.120	3.439	2.231e+07	5.329	66.7	
2P _{1/2} – 4D _{3/2}	460.4111	460.2821	651.1408	21719.720	1.931	1.935e+07	5.134	64.5	
2P _{3/2} – 3D _{5/2}	610.5334	610.3644	491.0337	16379.120	2.775	1.143e+06	10.980	100.0	
2P _{3/2} – 3D _{3/2}	610.5349	610.3659	491.0325	16379.080	2.266	1.143e+07	10.980	16.7	
2P _{1/2} – 3S _{1/2}	812.8456	812.6221	368.8185	12302.460	2.432	1.116e+07	5.329	33.3	
2P _{1/2} – 5D _{3/2}	413.3723	413.2557	725.2360	24191.270	1.123	9.040e+06	2.498	57.5	
2P _{3/2} – 4S _{1/2}	497.3130	497.1743	602.8245	20108.060	0.917	6.919e+06	2.840	38.8	
2P _{1/2} – 6D _{3/2}	391.6401	391.5292	765.4796	25533.650	0.767	4.964e+06	1.469	53.9	
2P _{3/2} – 4D _{5/2}	460.4179	460.2889	651.1312	21719.400	1.058	3.871e+06	5.134	77.4	
2P _{3/2} – 4D _{3/2}	460.4183	460.2893	651.1306	21719.380	0.864	3.871e+06	5.134	12.9	
2P _{1/2} – 4S _{1/2}	497.3046	497.1659	602.8347	20108.400	0.648	3.460e+06	2.840	19.4	
2P _{3/2} – 5S _{1/2}	427.4326	427.3124	701.3795	23395.500	0.494	3.170e+06	1.560	32.2	
2P _{1/2} – 7D _{3/2}	379.6098	379.5020	789.7385	26342.840	0.573	3.041e+06	0.941	52.0	
2P _{3/2} – 5D _{5/2}	413.3779	413.2613	725.2261	24190.940	0.615	1.810e+06	2.498	69.0	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
Γ/2π: natural linewidth of the upper level, 1/(2πτ). Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d ₇₀ ARC
2S _{1/2} → nP _{3/2}	230.602	230.317	230.186	230.072	230.011	229.953	0.0033
2P _{3/2} → nS _{1/2}	351.397	350.715	350.404	350.134	349.992	349.857	0.0060
2P _{3/2} → nD _{5/2}	351.356	350.698	350.395	350.131	349.991	349.857	0.0209
3S _{1/2} → nP _{3/2}	618.862	616.817	615.877	615.060	614.628	614.213	0.0052
3P _{3/2} → nS _{1/2}	804.085	800.528	798.905	797.506	796.770	796.068	0.0151
3P _{3/2} → nD _{5/2}	803.872	800.439	798.860	797.490	796.764	796.068	0.0418
3D _{5/2} → nP _{3/2}	827.699	824.046	822.367	820.912	820.142	819.404	0.0077
3D _{5/2} → nF _{7/2}	827.672	824.035	822.361	820.910	820.141	819.404	0.0449
4S _{1/2} → nP _{3/2}	1197.211	1189.582	1186.088	1183.064	1181.465	1179.933	0.0070
4P _{3/2} → nS _{1/2}	1450.888	1439.347	1434.111	1429.610	1427.245	1424.996	0.0276
4P _{3/2} → nD _{5/2}	1450.195	1439.061	1433.965	1429.558	1427.228	1424.996	0.0670
4D _{5/2} → nP _{3/2}	1483.369	1471.676	1466.332	1461.713	1459.272	1456.937	0.0184
4D _{5/2} → nF _{7/2}	1483.285	1471.641	1466.314	1461.706	1459.270	1456.937	0.0837
4F _{7/2} → nD _{5/2}	1484.786	1473.117	1467.778	1463.161	1460.720	1458.382	0.0080
5S _{1/2} → nP _{3/2}	1974.213	1953.555	1944.149	1936.037	1931.758	1927.668	0.0086

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d₇₀ = $\langle J||er||J' \rangle$ to *n* = 70 in ea₀ (calculated).
S → nP values come from a near-cancelling integral: treat them as an order of magnitude.

⁷Li hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2S _{1/2}	401.752043	-	F=1: -502.19 F=2: +301.31
2P _{1/2}	46.005(16)	-	F=1: -57.51 F=2: +34.50
2P _{3/2}	-3.050(16)	-0.22(3)	F=0: +11.16 F=1: +8.33 F=2: +2.45 F=3: -6.92
3S _{1/2}	93.105(7)	-	F=1: -116.38 F=2: +69.83
3P _{1/2}	13.50(20)	-	F=1: -16.88 F=2: +10.12
3P _{3/2}	-1.010(20)	-0.081(28)	F=0: +3.69 F=1: +2.76 F=2: +0.82 F=3: -2.29
3D _{3/2}	0.84(4)	-	F=0: -3.16 F=1: -2.32 F=2: -0.63 F=3: +1.90
3D _{5/2}	0.3436(10)	-	F=1: -1.80 F=2: -1.12 F=3: -0.09 F=4: +1.29
4S _{1/2}	35.0(4)	-	F=1: -43.81 F=2: +26.29
4P _{3/2}	-0.41(4)	-	F=0: +1.54 F=1: +1.13 F=2: +0.31 F=3: -0.92

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).